1.36 2D Matrices: Stretches and Enlargements

Worked Solutions

1 a)

Stretch, parallel to the x-axis, scale factor 2.

1 b)

Enlargement, centred at the origin, scale factor 7.

Alternative Method:

Stretch, parallel to the x-axis, scale factor 7, and a stretch, parallel to the y-axis, scale factor 7.

1 c)

Stretch, parallel to the y-axis, scale factor 5.

1 d)

Stretch, parallel to the x-axis, scale factor 10, and a stretch, parallel to the y-axis, scale factor 20.

1 e)

Enlargement, centred at the origin, scale factor $\frac{1}{3}$.

Alternative Method:

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$, and a stretch, parallel to the y-axis, scale factor $\frac{1}{3}$.

2 a)

$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

2 b)

$\mathbf{A}^2 = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$.

2 c)

$\mathbf{A}^3 = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$.

2 d)

$\mathbf{A}^{15} = \begin{pmatrix} 32768 & 0 \\ 0 & 32768 \end{pmatrix}$.

3)

The matrix represents a stretch, parallel to the y-axis, scale factor 3. So the image will be $3\times$ the area of the original shape. So the area of the shape's image is $3 \times 4 = 12$ units^2.

4)

The matrix represents an enlargement, centred at the origin, scale factor 3. So the image will be $3^2\times$ the area of the original shape. So the area of the shape's image is $3^2 \times 4 = 36$ units^2.

5)

$$\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} -2 & 0 & 4 \\ 0 & 9 & 0 \end{pmatrix} = \begin{pmatrix} -8 & 0 & 16 \\ 0 & 36 & 0 \end{pmatrix}$$

The coordinates of the vertices of triangle $T\prime$ are $(-8, 0)$, $(0, 36)$ and $(16, 0)$.

The area of the triangle is $= \frac{1}{2} \times 24 \times 36 = 432$ units^2.

6 a)

$\begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$.

6 b)

$\begin{pmatrix} 15 & 0 \\ 0 & 15 \end{pmatrix}$.

6 c)

$\begin{pmatrix} 1 & 0 \\ 0 & \frac{2}{3} \end{pmatrix}$.

6 d)

$\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$.

7 a)

$$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 2k \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$2k = 4 \Rightarrow k = 2$$

$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

7 b)

$$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 12 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 4k \\ -1 \end{pmatrix} = \begin{pmatrix} 12 \\ -1 \end{pmatrix}$$

$$4k = 12 \Rightarrow k = 3$$

$\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

7 c)

$$\begin{pmatrix}1 & 0\\0 & k\end{pmatrix}\begin{pmatrix}16\\9\end{pmatrix}=\begin{pmatrix}16\\3\end{pmatrix}\Rightarrow\begin{pmatrix}16\\9k\end{pmatrix}=\begin{pmatrix}16\\3\end{pmatrix}$$

$$9k=3\Rightarrow k=\frac{1}{3}$$

$$\begin{pmatrix}1 & 0\\0 & \frac{1}{3}\end{pmatrix}$$

7 d)

$$\begin{pmatrix}k & 0\\0 & 1\end{pmatrix}\begin{pmatrix}25\\-2\end{pmatrix}=\begin{pmatrix}5\\-2\end{pmatrix}\Rightarrow\begin{pmatrix}25k\\-2\end{pmatrix}=\begin{pmatrix}5\\-2\end{pmatrix}$$

$$25k=5\Rightarrow k=\frac{1}{5}$$

$$\begin{pmatrix}\frac{1}{5} & 0\\0 & 1\end{pmatrix}$$

7 e)

$$\begin{pmatrix}1 & 0\\0 & k\end{pmatrix}\begin{pmatrix}8\\-7\end{pmatrix}=\begin{pmatrix}8\\-3\end{pmatrix}\Rightarrow\begin{pmatrix}8\\-7k\end{pmatrix}=\begin{pmatrix}8\\-3\end{pmatrix}$$

$$-7k=-3\Rightarrow k=\frac{3}{7}$$

$$\begin{pmatrix}1 & 0\\0 & \frac{3}{7}\end{pmatrix}$$

7 f)

$$\begin{pmatrix}k & 0\\0 & 1\end{pmatrix}\begin{pmatrix}-30\\5\end{pmatrix}=\begin{pmatrix}-18\\5\end{pmatrix}\Rightarrow\begin{pmatrix}-30k\\5\end{pmatrix}=\begin{pmatrix}-18\\5\end{pmatrix}$$

$$-30k=-18\Rightarrow k=\frac{3}{5}$$

$$\begin{pmatrix}\frac{3}{5} & 0\\0 & 1\end{pmatrix}$$

8 a)

$$\begin{pmatrix}k & 0\\0 & k\end{pmatrix}\begin{pmatrix}2\\5\end{pmatrix}=\begin{pmatrix}8\\20\end{pmatrix}\Rightarrow\begin{pmatrix}2k\\5k\end{pmatrix}=\begin{pmatrix}8\\20\end{pmatrix}$$

$$2k=8\Rightarrow k=4\qquad 5k=20\Rightarrow k=4$$

$$\begin{pmatrix}4 & 0\\0 & 4\end{pmatrix}$$

8 b)

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} -240 \\ 90 \end{pmatrix} = \begin{pmatrix} -48 \\ 18 \end{pmatrix} \Rightarrow \begin{pmatrix} -240k \\ 90k \end{pmatrix} = \begin{pmatrix} -48 \\ 18 \end{pmatrix}$$

$$-240k = -48 \Rightarrow k = \frac{1}{5} \qquad 90k = 18 \Rightarrow k = \frac{1}{5}$$

$\begin{pmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$

8 c)

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 20 \\ 600 \end{pmatrix} = \begin{pmatrix} 64 \\ 1920 \end{pmatrix} \Rightarrow \begin{pmatrix} 20k \\ 600k \end{pmatrix} = \begin{pmatrix} 64 \\ 1920 \end{pmatrix}$$

$$20k = 64 \Rightarrow k = \frac{16}{5} \qquad 600k = 1920 \Rightarrow k = \frac{16}{5}$$

$\begin{pmatrix} \frac{16}{5} & 0 \\ 0 & \frac{16}{5} \end{pmatrix}$

8 d)

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} -360 \\ -440 \end{pmatrix} = \begin{pmatrix} -342 \\ -418 \end{pmatrix} \Rightarrow \begin{pmatrix} -360k \\ -440k \end{pmatrix} = \begin{pmatrix} -342 \\ -418 \end{pmatrix}$$

$$-360k = -342 \Rightarrow k = \frac{19}{20} \qquad -440k = -418 \Rightarrow k = \frac{19}{20}$$

$\begin{pmatrix} \frac{19}{20} & 0 \\ 0 & \frac{19}{20} \end{pmatrix}$

9 a)

Area of $T = \frac{1}{2} \times (8-4) \times (9-3) = \frac{1}{2} \times 4 \times 6 = 12$ units^2.

9 b)

$$\begin{pmatrix} 5 & 0 \\ 0 & 12 \end{pmatrix}\begin{pmatrix} 4 & 8 & 7 \\ 3 & 3 & 9 \end{pmatrix} = \begin{pmatrix} 20 & 40 & 35 \\ 36 & 36 & 108 \end{pmatrix}$$

So $A\prime(20, 36)$, $B\prime(40, 36)$ and $C\prime(35, 108)$.

Area of $V = \frac{1}{2} \times (40-20) \times (108-36) = \frac{1}{2} \times 20 \times 72 = 720$ units^2.

Alternative Method (using determinants):

$\begin{vmatrix} 5 & 0 \\ 0 & 12 \end{vmatrix} = 5 \times 12 - 0 \times 0 = 60.$

Area of $V = 60\times$ Area of $T = 60 \times 12 = 720$ units^2.

10)

Area of $K = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 2 \times 5 = 7$.

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 & 1 \\ -5 & 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -a & 0 & a \\ -5a & 0 & 2a & 0 \end{pmatrix}$$

Area of $K\prime = \frac{1}{2} \times 2a \times 2a + \frac{1}{2} \times 2a \times 5a = 7a^2$.

$$7a^2 = 2 \times 7 \Rightarrow a^2 = 2 \Rightarrow a = \sqrt{2}$$

So $K\prime$ has coordinates $(0, -5\sqrt{2})$, $(-\sqrt{2}, 0)$, $(0, 2\sqrt{2})$ and $(\sqrt{2}, 0)$.

11 a)

$\begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix}$.

11 b)

$$\begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2m + n & -m \end{pmatrix} = \begin{pmatrix} 18n + 1 \\ m - 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 14m + 7n \\ 3n - 3m \end{pmatrix} = \begin{pmatrix} 18n + 1 \\ m - 1 \end{pmatrix}$$

$14m + 7n = 18n + 1 \Rightarrow 14m - 11n = 1 \quad (1)$

$3n - 3m = m - 1 \Rightarrow 4m - 3n = 1 \quad (2)$

$\therefore m = 4$ and $n = 5$.

12)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} -2 \\ 8 \end{pmatrix} - \begin{pmatrix} 2 \\ -6 \end{pmatrix} = \begin{pmatrix} -4 \\ 14 \end{pmatrix}$.

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -4 \\ 14 \end{pmatrix} = \begin{pmatrix} -12 \\ 42 \end{pmatrix}$$

$\overrightarrow{OA\prime} = \overrightarrow{OB} + \begin{pmatrix} -12 \\ 42 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \end{pmatrix} + \begin{pmatrix} -12 \\ 42 \end{pmatrix} = \begin{pmatrix} -10 \\ 36 \end{pmatrix}$.

So $A\prime(-10, 36)$.

13)

$\overrightarrow{BA\prime} = \overrightarrow{OA\prime} - \overrightarrow{OB} = \begin{pmatrix} 10 \\ -1 \end{pmatrix} - \begin{pmatrix} -8 \\ -3 \end{pmatrix} = \begin{pmatrix} 18 \\ 2 \end{pmatrix}$.

$$\begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix} = \begin{pmatrix} 180 \\ 20 \end{pmatrix}$$

$\overrightarrow{OA} = \overrightarrow{OB} + \begin{pmatrix} 180 \\ 20 \end{pmatrix} = \begin{pmatrix} -8 \\ -3 \end{pmatrix} + \begin{pmatrix} 180 \\ 20 \end{pmatrix} = \begin{pmatrix} 172 \\ 17 \end{pmatrix}$.

So $A(172, 17)$.